

Technical Report N° 05 — Acquisition of the Qualitative Database

Project: *Forecasting Bitcoin Prices and Analyzing Geopolitical Risks: A Residual Analysis Approach using LSTM (Geo-BTC).*

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Phase: Qualitative research (Phase 2).

Status: Closed.

Companion artefacts: data/external/qualitative_analysis.csv,
data/external/qualitative_analysis.xlsx.

1 · Purpose

This report documents the **research process** behind the qualitative event database used in Phase 2 and presents a **flag-by-flag overview** of the eight event windows that explain the LSTM residual outliers in 2015. It is the human-readable counterpart of the machine-readable artefacts under `data/external/`.

2 · Research design

2.1 · Triggering criterion

The qualitative team was activated after Phase 1 produced `investigation\_periods.csv`. For every outlier detected by the $\pm 2\sigma$ rule, an asymmetric $[-3, +7]$ -day investigation window was opened. The post-event tail (+7) reflects the empirical lag observed between a geopolitical event and its market footprint.

2.2 · Source corpus

A **convenience-then-replication** sampling procedure was adopted. For each investigation window the analyst:

1. **Convenience sweep** — looked up the equivalent dates on five primary outlets:

- **CoinDesk Bitcoin Price Index (BPI)** for daily price drivers.
- **CNN Markets** and **BBC News** for macroeconomic context.
- **Wall Street Journal** for institutional / regulatory angle.
- **Bitcoin.org** for exchange-level volume and liquidity.

2. **Replication** — discarded any candidate event that could not be independently corroborated by at least two of the five sources.

3. **Window expansion** — when the literal `[-3, +7]` window missed a clearly causal event by 2-3 days, the window was relaxed; the audit trail is preserved in the *Date* column of `qualitative_analysis.csv`.

2.3 · Coding scheme

Each retained event was annotated with five fields:

Field	Description
Date	Free-form bilingual period string (PT-BR / EN).
Fluctuation	Direction of the price move during the window — <code>Up / Down</code> .
Category	One of eight categories that emerged inductively (see §3).
Event (Macro)	One- or two-sentence description of the macro narrative.
Breakdown (Micro)	Detailed breakdown of the proximate market mechanism.
Price Movement	Open / close prices and percentage during the window.

Disagreements between analysts were resolved by majority vote among the three authors; ties (none observed) would have been broken by reverting to the most conservative source.

3 · Inductive category taxonomy

Eight categories emerged from the coding. They are partitioned across the two behavioural regimes (Down and Up):

Regime	Categories
--------	------------

Down (negative shock)	Ecosystem Vulnerability · Global Macroeconomic Shock
Up (positive shock)	Institutional Interest · Chinese Market Demand Surge · Global Expansion · Technical Consolidation · Speculative Re-entry · Parabolic Move — FOMO

The down-regime is dominated by **exogenous** shocks (hacks, currency devaluations) whereas the up-regime is dominated by **endogenous** market dynamics (volume surges, leverage cascades, FOMO loops). This asymmetry is itself a finding worth reporting in the paper.

4 · Flag-by-flag overview

The table below summarises each of the eight event windows in chronological order. Numerical price movements are reproduced verbatim from the qualitative dataframe.

#	Window	Regime	Category	Event (Macro)	Price Movement
1	15-25 Mar 2015	Down	Ecosystem Vulnerability	Failed attempt to break US\$ 300, Coinapult and AllCrypt hacks revive Mt. Gox trauma.	US\$ 286 → US\$ 245 (-14%)
2	09-19 Jul 2015	Up	Institutional Interest	Banks and funds shift from observation to active blockchain study; first concrete institutional signal.	US\$ 260 → US\$ 290 (+10 to +12%)

3	15-25 Aug 2015	Down	Global Macroeconomic Shock	China devalues the yuan three times in a week; Shanghai falls 8% on Aug-24 ("Black Monday of 2015").	US\$ 265 → US\$ 210-220 (-18 to -20%)
4	27 Oct - 06 Nov 2015	Up	Chinese Market Demand Surge	Volume on Huobi and OKCoin explodes (78% of global volume). BTC breaks US\$ 300 for the first time in months.	US\$ 280 → US\$ 380+ (+30 to +35%)
5	30 Oct - 09 Nov 2015	Up	Global Expansion	Move stops being only Chinese: BTC China triples volume, Bitfinex doubles, BTC-e +128%, Bitstamp +117%.	US\$ 300 → US\$ 480+ (+25 to +30%)
6	01-11 Nov 2015	Up	Technical Consolidation	Digestion phase; old resistances at US\$ 320 and US\$ 350 become support; Nov-26 sees record 2.86M BTC traded.	US\$ 325 → US\$ 380 (+15 to +17%)

7	23 Nov - 03 Dec 2015	Up	Speculative Re-entry	Mid-November correction opens room for fresh speculative capital; FX-protection narrative for Chinese investors persists.	US\$ 325 → US\$ 360 (+10 to +12%)
8	05-15 Dec 2015	Up	Parabolic Move — FOMO	Yearly peak; BTC breaks US\$ 380, US\$ 400 and US\$ 450 in sequence; CoinDesk publishes the "From Worst to First" retrospective.	US\$ 360 → US\$ 450 (+20 to +25%)

5 · Aggregate-flag panorama

The eight windows partition the year into a small number of narratives:

- **Two single-flag down-windows** (March + August) cover the only significant negative shocks of 2015 and concentrate ~30% of the total absolute residual mass.
- **Six up-windows**, all clustered between July and December, document the asymmetric rally that took BTC from ~US\$ 210 in August to ~US\$ 450 in December.
- **Five of the eight windows are China-centred** (yuan devaluation, Huobi/OKCoin volume surge, Chinese exchange expansion, FX-protection narrative, MMM-Ponzi tail), confirming that 2015 is the year in which Chinese exchange dynamics dominated global price discovery.
- The two pure **endogenous** categories — Technical Consolidation and Parabolic Move — FOMO — show the highest per-flag MAE values, consistent with the hypothesis that LSTM forecasts struggle most in regimes driven by reflexivity rather than information shocks.

6 · Reproducibility

The full machine-readable database is shipped under

```
`data/external/qualitative_analysis.csv` (UTF-8, RFC 4180) and  
`data/external/qualitative_analysis.xlsx` (formatted, with column  
widths and frozen header). Both files are version-controlled and  
referenced by the merge step in  
`src/inference/merge.py` via the canonical paths declared in  
`configs/paths.py` (QUALITATIVE_CSV_PATH,  
QUALITATIVE_XLSX_PATH).
```

To reproduce the merge → ER → storyboard pipeline that consumes this database, run from the repository root:

```
make merge    # build inference_table.csv  
make er       # compute the Explaining Ratio  
make plot     # render the inference storyboard (HTML + PNG)  
# or simply  
make phase2
```

7 · Limitations and next steps

- **Sample size.** Eight windows is small in absolute terms; statistical power for category-level conclusions is limited. The Phase-2 design is qualitative-explanatory, not inferential.
- **Source bias.** The five-source corpus is heavily Anglo-American; CN-language sources (Sina Finance, Caixin) were not systematically scanned. A follow-up effort could replicate the procedure on a Mandarin corpus.
- **Causal direction.** Containment-based matching does not establish causality — only co-occurrence. The paper presents the Explaining Ratio as descriptive evidence, not a counterfactual claim.

8 · Closing

Together with **Technical Report N° 04**, this document closes the inference-and-analysis phase of Geo-BTC. The pipeline now has a single

reproducible entry point (`make pipeline`), a self-contained qualitative database, and three artefacts that feed the manuscript directly: `inference_table.csv`, `explaining_ratio.csv` and `inference_storyboard_2015.html`.